Downstream Benchmarks of Pareto-Optimal Compressed Models: Qwen2.5-0.5B-Instruct with UBQ-ECSQ

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Abstract

A multi-objective evolutionary search (NSGA-II) over global quantization and pruning parameters produced a Pareto front of 100 compressed versions of Qwen2.5-0.5B-Instruct, trading calibration loss against encoded model size (239–495 MB, against 988 MB in bf16). Solutions at calibration-loss increases of 0%, 1%, 2%, 5% and 10% over the uncompressed model are evaluated, together with the uncompressed model, on eight standard benchmarks: six zero-shot commonsense and reasoning tasks (PIQA, ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, BoolQ), 5-shot MMLU and 5-shot GSM8K. At 8.0 bits per weight the model is 2.00× smaller with no loss of accuracy. At 6.0–6.7 bits per weight (2.39–2.66× smaller) it keeps 98% of the average accuracy. At 5.0 bits per weight (3.2× smaller) it keeps 88%, with most of the loss on GSM8K and BoolQ.

1 Compression scheme

All 169 weight matrices of the model are compressed, including the token embedding, which is tied to the output head (494.0M weights, 100% of the parameters). Every matrix uses the same five parameters (genes). For a weight w in output row r, with σ_r and $m_r = \max|w_r|$ taken from the original weights:

1. **Pruning:** $w \to 0$ if $\alpha\,\sigma_r \le w \le \beta\,\sigma_r$, with $\alpha \le 0 \le \beta$.
2. **Uniform quantization (UBQ):** a symmetric, zero-preserving grid of k levels with step $s_r = c\,m_r/\lfloor (k-1)/2 \rfloor$, clipping factor $c \in [0.7, 1]$; pruned weights stay 0.
3. **Entropy-constrained merge (ECSQ):** each non-empty level gets a model-wide centroid; contiguous levels are merged by dynamic programming to minimize $D + \lambda R$ (added squared error plus entropy of level occupancy), keeping a fraction ρ of the levels. The zero level is never merged. $\rho = 1$ means no merge.

Encoded size is the smaller of a dense code ($\lceil \log_2 k_{\text{used}} \rceil$ bits per weight) and a bitmap code (1 bit per weight plus a code per non-zero), plus a 16-bit scale per row and a lookup table per matrix. Sizes are in MB (10^6 bytes); bits per weight include scales and tables.

2 Search

- **Objectives (minimized):** $\Delta\mathcal{L}$, the next-token cross-entropy on the calibration set minus that of the uncompressed model, and the encoded size.
- **Constraint:** $\Delta\mathcal{L} \le 1.0$ nats.

- **Calibration set:** 8 sequences of 2048 tokens from C4 (train, English), fixed for the whole run. Uncompressed loss $\mathcal{L}_0 = 2.8854$ nats.
- **Genes:** $\log_2 k$ with $k \in [16, 256]$ (range set by a warm-up on the model's error-vs-k curve), $c \in [0.7, 1]$, $\alpha \in [-1, 0]$, $\beta \in [0, 1]$, $\rho \in [0, 1]$.
- **NSGA-II:** population 100, 50 generations, seed 42.

3 Pareto fronts

Figure 1 shows the final front on the objective of the search, the calibration loss. Figure 2 scores the same 100 solutions on held-out text: perplexity on the full WikiText-2 test split and on 256 windows of C4 validation, with 2048-token windows. The test fronts follow the calibration front: the loss stays near the uncompressed model's down to about 370 MB and rises steeply below 310 MB.

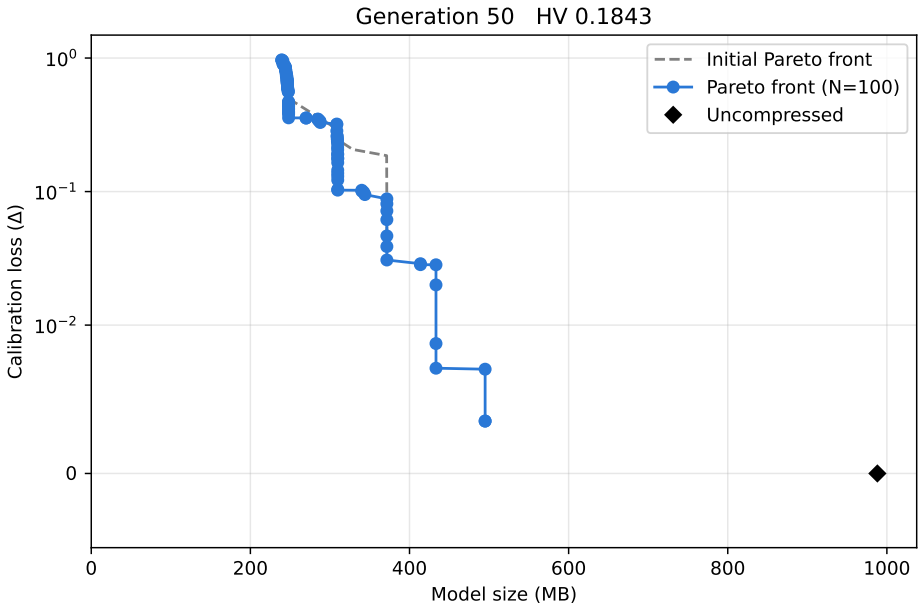
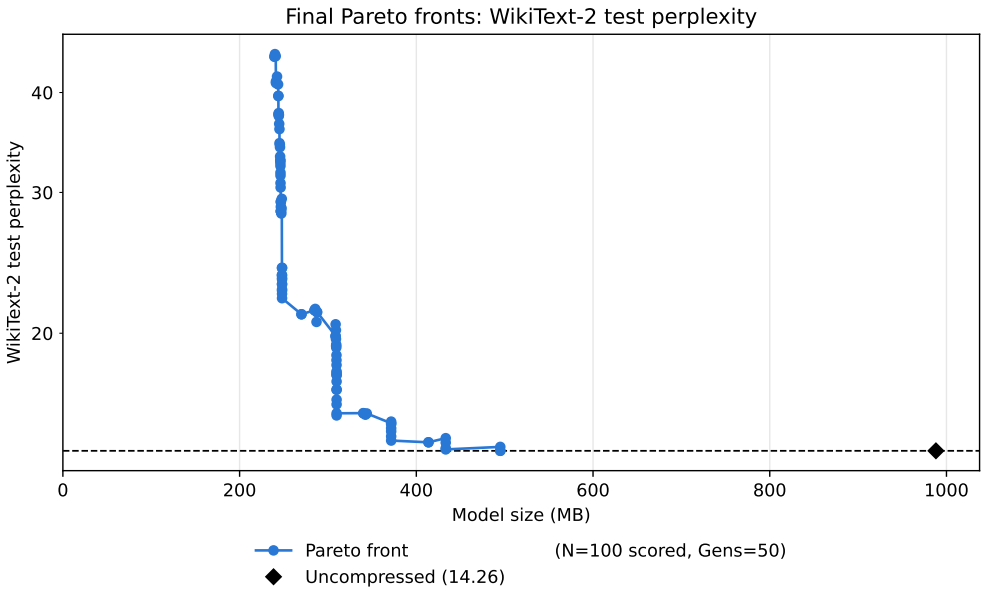
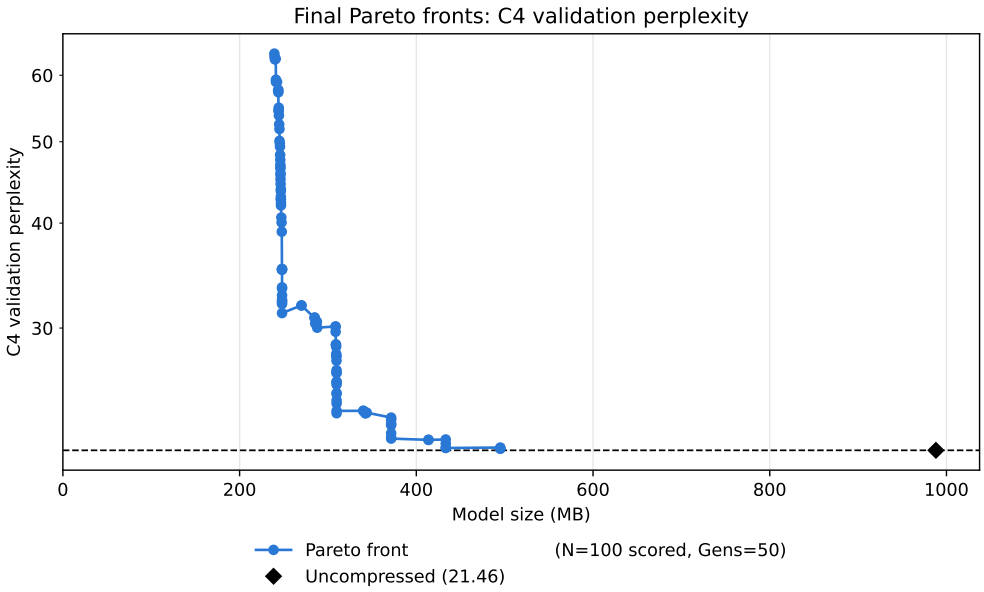


Figure 1: Final Pareto front (100 solutions) of calibration $\Delta\mathcal{L}$ against encoded size, with the front of the initial population and the uncompressed model.



(a) WikiText-2 test



(b) C4 validation

Figure 2: Test perplexity of the final Pareto front against encoded size. The dashed line is the uncompressed model.

4 Selected models

The calibration error of a solution is $\Delta\mathcal{L}/\mathcal{L}_0$. For each level (0, 1, 2, 5, 10%) the smallest solution within it is selected. No solution has zero error, so the most accurate one (0.12%) stands for the 0% level. Every solution smaller than 309 MB exceeds 10% error, so the 5% and 10% models are nearly the same size but reach it with different settings: Q10 prunes more ($\beta = 0.32$ against 0.02) and clips harder ($c = 0.85$ against 0.93).

Model	k	c	Sparsity (%)	Calib. err. (%)	Size (MB)	Smaller by	Bits/w	Huffman bits/w	WikiText-2 ppl	C4 ppl
Uncompressed (bf16)	–	–	0.0	0.00	988	1.00×	16.00	–	14.26	21.46
Q0 (0%)	164	0.995	2.2	0.12	495	2.00×	8.02	6.47	14.26	21.55
Q1 (1%)	65	0.934	5.3	1.00	414	2.39×	6.70	5.23	14.61	22.08
Q2 (2%)	52	0.949	6.8	1.61	372	2.66×	6.01	4.84	15.08	22.49
Q5 (5%)	29	0.928	11.8	4.75	310	3.19×	5.01	4.05	17.00	25.05
Q10 (10%)	28	0.849	19.2	9.87	309	3.20×	5.00	3.98	20.52	29.70

Table 1: Selected solutions. Sparsity counts pruned weights and weights rounded to zero. Huffman bits/w is the storage with Huffman-coded codes (reported only, not optimized). None of the five merges levels ($\rho = 1$).

5 Benchmark protocol

All models are evaluated with the same harness (lm-evaluation-harness 0.4.13), in bf16, on full evaluation splits, without a chat template, with fixed seeds. Compressed weights are reconstructed in place, so the models differ only in their weight values.

Benchmark	Split	Questions	Shots	Metric
PIQA	validation	1,838	0	accuracy, length-normalized
ARC-Easy	test	2,376	0	accuracy, length-normalized
ARC-Challenge	test	1,172	0	accuracy, length-normalized
HellaSwag	validation	10,042	0	accuracy, length-normalized
WinoGrande	validation	1,267	0	accuracy
BoolQ	validation	3,270	0	accuracy
MMLU	test	14,042	5	accuracy (57 subjects)
GSM8K	test	1,319	5	exact match of the final number

Table 2: Benchmarks. Multiple-choice tasks pick the answer with the highest log-likelihood; GSM8K generates a solution and compares the extracted final number with the reference.

6 Results

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU 5-shot	GSM8K 5-shot	Average
Uncompressed	988	70.4	59.2	33.1	52.4	56.0	67.8	47.0	31.2	52.1
Q0 (0%)	495	70.2	59.9	33.5	52.4	56.6	68.4	47.0	31.7	52.5
Q1 (1%)	414	70.2	59.6	33.7	52.5	55.9	64.3	46.5	27.6	51.3
Q2 (2%)	372	69.5	61.0	32.8	52.0	54.2	66.6	46.1	26.7	51.1
Q5 (5%)	310	69.2	55.2	32.0	50.0	56.4	49.7	45.0	9.7	45.9
Q10 (10%)	309	67.5	56.2	31.0	47.7	57.2	56.1	38.3	12.6	45.8
Std. error		1.1	1.0	1.4	0.5	1.4	0.8	0.4	1.3	

Table 3: Accuracy (%, higher is better), with the unweighted average over the eight benchmarks. Last row: standard error of the uncompressed model's accuracy.

Model	Size (MB)	PIQA	ARC-e	ARC-c	HellaS.	WinoG.	BoolQ	MMLU	GSM8K	Avg. kept (%)
Q0 (0%)	495	−0.2	+0.7	+0.4	−0.1	+0.6	+0.6	0.0	+0.5	100.6
Q1 (1%)	414	−0.2	+0.5	+0.6	0.0	−0.2	−3.5	−0.4	−3.6	98.4
Q2 (2%)	372	−0.9	+1.9	−0.3	−0.5	−1.8	−1.2	−0.8	−4.5	98.0
Q5 (5%)	310	−1.2	−4.0	−1.1	−2.4	+0.3	−18.1	−2.0	−21.5	88.0
Q10 (10%)	309	−2.9	−2.9	−2.1	−4.7	+1.2	−11.8	−8.6	−18.6	87.9

Table 4: Change in accuracy against the uncompressed model (percentage points; negative is a loss). Avg. kept is the compressed model's average accuracy as a percentage of the uncompressed model's.

Observations.

- **8 bits per weight is lossless.** Q0 is 2.00× smaller and every benchmark is within 0.7 points of the uncompressed model, inside the standard errors (100.6% kept).
- **1–2% calibration error keeps 98% of the accuracy.** Q1 and Q2 are 2.39× and 2.66× smaller. The commonsense tasks and MMLU stay within 2 points; GSM8K drops 3.6–4.5 points and BoolQ up to 3.5.
- **At 5.0 bits per weight, about 12% of the average accuracy is lost.** Q5 and Q10 are 3.2× smaller and keep 88%. The commonsense tasks lose at most 5 points; BoolQ breaks (Q5: 49.7%, below the 62.2% of always answering "yes").
- **GSM8K is the most sensitive benchmark:** 31.2% uncompressed, 27–28% at 1–2% error, 10–13% at 5–10% error.
- **Calibration loss does not rank the models downstream.** Q5 has half the calibration error of Q10 and lower test perplexity, and is better on MMLU (45.0 against 38.3), but worse on BoolQ and GSM8K; the two averages are equal. Below about 6 bits per weight, models need a benchmark check before use.